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HMS-IDS: Threat Intelligence Integration for Zero-Day Exploits and Advanced Persistent Threats in IIoT

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Abstract

Critical Industries such as Manufacturing, Power, and Intelligent Transportation are increasingly using IIoT systems, making them more susceptible to cyberattacks. To counter these cyberattacks, policymakers have made strong guidelines, and various security provisions like secure authentication and encryption mechanisms as effective countermeasures for these systems. The exponential rise in cyberattacks has proven that all these measures are not sufficient to protect IIoT systems and have certain limitations. Considering the progress in Artificial Intelligence, it is widely acknowledged that Machine Learning (ML) based Intrusion Detection Systems (IDS) hold significant potential for identifying these cyberattacks. Numerous ML-based IDS have been proposed, which are capable of detecting known attacks but do not perform well in recognizing the "Unknown-Attacks" or Zero-Day Attacks (ZDAs) and Advanced Persistent Threats (APTs); hence, one of the most prominent concerns in the cyber industry is how threat intelligence could be used to protect against these exploits. The proposed "Hybrid Multi-Stage Intrusion Detection System" (HMS-IDS) is driven by supervised and unsupervised approaches to identify both known and unknown cyber-attacks in IIoT environments. By carefully evaluating the esteemed CIC-ToN-IoT dataset, the proposed IDS model attains staggering levels of accuracy, reaching an impressive 99.49% in detecting known attacks and an exceptional 98.936% in identifying unknown attacks. These compelling findings unequivocally substantiate the system's efficacy in real-time detection of malicious cyber incursions targeting IIoT devices, thereby underscoring its tremendous potential for wide-scale implementation and practical deployment. To validate the proposed model's reliability, the performance evaluation is also performed on state-of-the-art datasets, namely KDD-99 Cup, NSL-KDD, CICIDS 2017.

Keywords IIoT · Intrusion Detection System · CIC-ToN-IoT · Cyber-attack · Zero-Day Attacks

1 Introduction

Industries have continually evolved, aiming for more efficient, productive, manageable, quality-oriented, and secure operations [1]. The advancement of Information Technology (IT), significantly impacted the Industrial Control Systems (ICSs), specifically the Supervisory Control and Data Acquisition (SCADA) systems, which enable the remote management of complex production lines through Human–Machine Interfaces (HMIs) [2]. The expansion of the Internet of Things (IoTs) has evolved industries, referred to as the Industrial Internet of Things (IIoTs), which is also termed as the fourth generation of SCADA systems [3]. This revolution opens the path to meet the ever-growing demand for more efficient, interoperable, manageable, and reduced

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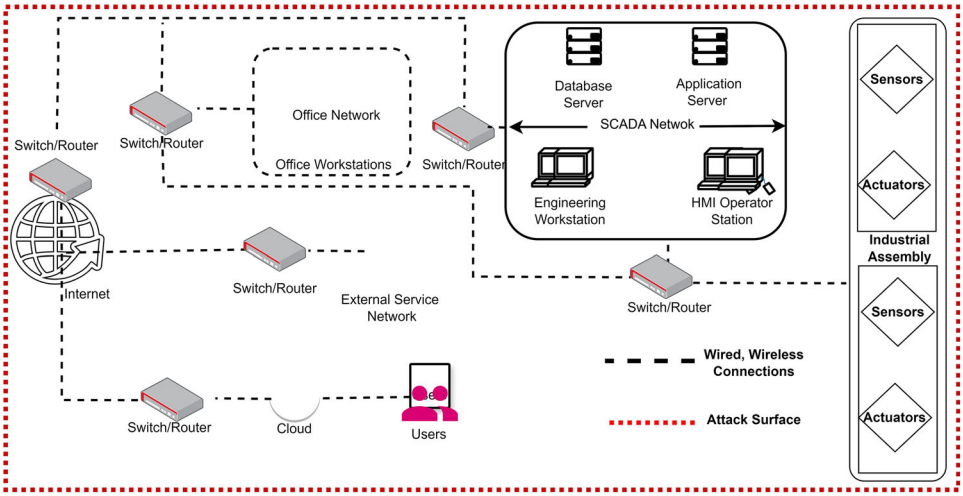


Fig. 1 In an Industrial Internet of Things (IIoT) scenario, attack surfaces can be quite extensive. Here is an Industrial IoT scenario of the potential attack surfaces to consider

cost of production [4]. The consequences of this upgrade lead to an increase in security issues due to the significant expansion in their vulnerability to cyberattacks [5] (Fig. 1). SCADA systems, which are currently in use, are widely adopted in vital sectors like manufacturing, electricity grid operations, wastewater treatment, intelligent public transport, and smart city operations, by utilizing the advancements in IIoT technologies [6]. In reality, the prevalence of hacks leveraging IIoT technologies has witnessed a significant increase within the last decade [7]. Advanced Persistent Threats (APTs) are highly sophisticated methodologies frequently employed in the context of these cyber intrusions. The term "Advanced" pertains to the utilization of intricate attack techniques by the assailants. These techniques encompass the employment of publicly accessible exploits that specifically target widely recognized vulnerabilities, as well as the deployment of custom payloads and delivery mechanisms, commonly known as zero-day exploits [8]. The later are designed to adapt and modify based on the response of the targeted system. "Persistence" refers to the continuous and uninterrupted communication between the attacker and the vulnerable systems until the desired objective of the attack is successfully achieved. In brief, APTs encompass precisely orchestrated, covert assaults that leverage sophisticated exploitation methodologies to target specific entities, to maintain prolonged persistence.

To detect these ZDAs and APTs, Intrusion detection systems (IDSs) have been played a significant role for a long time [9]. The developments in cybersecurity have witnessed the integration of ML methodologies, enabling the acquisition of knowledge from data through statistical algorithms. ML models exhibit proficiency in anticipating, categorizing, and forecasting events and scenarios by discerning intricate data patterns that may elude domain experts [10] and provide the self-healing [11] in case of cyberattacks. This has led to the adoption of ML in fortifying organizational security across various industries. Traditional "Signature-based IDS" employs a scanning mechanism to detect recognizable attacks by identifying specific Indicators of Compromise (IOCs). However, these systems face challenges in detecting zero-day attacks, which exploit unknown vulnerabilities before they are addressed, posing a substantial security risk [12]. In contrast, ML based IDSs [13–16] leverage behavioral patterns learned from incoming network packets to identify anomalies or malicious content, going beyond IOCs. These models possess greater potential for detecting zero-day attacks, as they learn intricate traffic patterns and do not rely solely on established IOCs.

In the IIoT environment, the proposed work focuses on the integration of threat intelligence for zero-day attacks and advanced persistent threats [17]. An IIoT ecosystem incorporates linked devices along with critical infrastructure, thus pose special security issues. The study suggests an intelligent IDS that is especially made to recognize both known and zero-day cyberattacks in IIoT systems. The suggested IDS improves the efficiency of threat detection in IIoT environments by incorporating machine learning methods and adopting a tiered ensemble structure. By identifying complex patterns and indicators of malicious behavior, the system overcomes the drawbacks of standard signature-based techniques [18]. Data preparation and feature engineering approaches are also employed to optimize the identification of threats. The integration of K-Means clustering offers a flexible solution to the shortcomings of conventional systems in identifying zero-day threats. In order to determine if a packet belongs to a known attack, an unknown assault, or benign activity, the study offers a two-step verification process for packet detection in the IIoT context. The ability to learn from data is made possible by machine learning algorithms, which empower the IDS to recognize and understand

complicated data patterns that frequently escape domain specialists. By leveraging machine learning algorithms and analyzing network behaviors, these systems enhance detection capabilities, enabling organizations to provide effective safegaurd from the evolving cyberatacks [19]. The research paper contributes to the field of threat intelligence integration for IIoT by proposing an advanced IDS architecture that strengthens the resilience and effectiveness of security measures in safeguarding critical IIoT systems.

In order to facilitate a thorough comprehension of the research work, the background and motivation are expounded in Sect. 1. Section 2 deals with the Related works, where a few notable works on known and unknown attack detection have been done. Section 3 explains Dataset details and Sect. 4 deals with the proposed methodology, which introduces an attack path assessment framework and also gives the overview about Runtime complexity of the implementation of the proposed threat detection model. Results and observations are discussed in Sect. 5 whereas Sect. 6 serves as the concluding segment of the paper, wherein it also delves into potential avenues for further research possibilities.

2 Related Works

This section thoroughly discusses existing research works on IDS focusing on identification and mitigations of zero-day threats. Most research papers emphasize known attack detection using ML techniques, with only a few addressing ML-based IDSs for zero-day attacks [20]. Anomaly detection systems with unsupervised models classify all attacks as unknown, but may suffer from false alarms and alert fatigue due to neglecting attack behavior patterns [21].

Zhang et al. [22] looked at how well IDSs based on ML could find zero-day attacks. Researchers used Zero-Shot learning (ZSL) to model attacks that had never been seen before. To finish this task, an advanced computing model called a "sparse autoencoder" was used. With this idea, a semantic space was made for attack characteristics that could be recognized. The IDS found new attack trends by figuring out what the different parts of an attack meant. The average attack accuracy in the collection was 88.3%, which is a good sign. The aforementioned discoveries illuminate the potential efficacy of IDSs based on machine learning in accurately detecting and mitigating zero-day attacks. The incorporation of Zero-Shot learning (ZSL) methodologies and semantic mapping has significantly contributed to the improvement of IDSs, particularly in relation to the identification of ZDAs. This research offers significant insights into the utilization of sophisticated techniques that enhance the performance of IDS in identifying and mitigating emerging threats. Thus, this paper discusses for the novel opportunities to dive deeper in the IDS and propel their progress.

Hindy et al. [23] performed an assessment aimed at enhancing unsupervised outlier-based detection systems for the purpose of ZDA detection. The methodology employed, utilizing autoencoders, was designed to mitigate the occurrence of erroneous alerts and improve the precision of detection. Using the CICIDS2017 and NSL-KDD datasets, the study reported that the accuracy of the detection of ZDA ranges from 75 to 99%. Although the models did not explicitly consider attack behaviors and the evaluation lacked measurements of undetected attacks and false alarms, the findings demonstrate the potential of the autoencoder-based technique. Further research is needed to address limitations and assess performance in real-world scenarios.

In [24], the authors introduced a pioneering methodology to proficiently discern ZDAs by assimilating attribute learning into an IDS. The proffered framework aimed to transcend the constraints of extant anomaly detection methodologies by amalgamating a Random Forest (RF)-based technique for selecting salient features with a spatial clustering approach to facilitate attribute conversion. By conducting rigorous empirical research, the investigation demonstrated the impressive effectiveness of the proposed framework, surpassing previous cutting-edge methods in the field of anomaly detection. The Attribute learning framework utilized an unsupervised approach to transform network data samples into cluster attributes, thereby facilitating the extraction of cogent and illustrative characteristics for subsequent analysis and classification. The evaluation of the framework, utilizing the NSL-KDD dataset, yielded highly auspicious results, manifesting an impressive accuracy of 34.71% in correctly identifying Denial of Service (DoS) attacks (specifically apache2) and Probe attacks (specifically saint). In stark contrast, a decision tree classifier demonstrated a notably inferior accuracy of a mere 13.59%. These findings underscore the resounding efficacy and inherent potential of the proposed framework in successfully detecting zero-day attacks, thereby showcasing its incontrovertible superiority over conventional methodologies prevalent in the field of IDS.

Kumar et al. [25] unveiled an intricately crafted model purposefully engineered to detect high-volume zero-day attacks. The model aptly confronts the challenge of discerning and classifying diverse attack categories based on the distinctive traffic volume attributes, thereby augmenting the overarching detection capacity and precision of the IDS. Comprising two distinctive stages, namely signature generation and evaluation, the model intricately weaves a tapestry of mechanisms for optimal performance. In the signature generation phase, potent low-volume attacks, known as heavy-hitters, form the bedrock for identifying the formidable high-traffic volume attacks, while the ingenious graph technique is adroitly harnessed to derive signatures for low-volume attacks, thus enabling their prompt detection and methodical categorization. Rigorous performance

evaluation of the model is executed leveraging both real-time attack datasets and the venerable CICIDS18 dataset as benchmarks. The resultant outcomes bear testament to the model's unparalleled prowess, showcasing exceptional accuracies of 91.33% in binary and 90.35% in multi classification of attacks, when tested against the real-time attack dataset. Furthermore, when pitted against the esteemed CICIDS18 dataset, the model demonstrates remarkable detection accuracies of 91.62% and 88.98%. These resounding findings emphatically underscore the model's profound efficacy in accurately identifying and categorizing zero-day attacks, with a particular focus on those characterized by substantial traffic volumes. Given the complexity and evolving nature of the cybersecurity landscape, further concerted endeavors in the form of comprehensive research and exhaustive evaluation are indispensable to validate the model's performance across diverse attack scenarios and real-world environments, thereby establishing its unassailable robustness and unwavering reliability.

Sabeel et al. [26] advanced a formidable model tailored to identify high-volume zero-day attacks within IDS. The framework ingeniously amalgamates heavy-hitter low-volume attacks with a graph technique to deftly discern and categorize attacks based on the distinctive attributes of traffic volume. The model demonstrated remarkable prowess, achieving soaring levels of detection accuracy ranging from 91.33% to an impressive 91.62% when subjected to real-time datasets as well as esteemed benchmark datasets. Moreover, the investigation entailed evaluating artificial intelligence (AI) models against non-conventional attacks, where the S-DTC model outshone its counterparts, namely the DNN and L-SVC models. The research underscored the dire necessity for thorough evaluations of IDS models when confronted with anomalous and unfamiliar attacks, shining a spotlight on pivotal factors such as training dataset size and the meticulous selection of features. The future trajectory of this inquiry will delve into exploring adversarial semi-supervised methodologies while extending the evaluation to encompass a diverse array of attack datasets, thus fortifying the depth and breadth of the study's findings.

Mokhtari et al. [27] embarked upon an exploration into the performance of a distinctive Measurement Intrusion Detection System (MIDS) for unearthing clandestine attacks within SCADA systems. The MIDS, relying on the discernment of measurement data, exhibited unparalleled proficiency in detecting surreptitious behaviors when contrasted against the conventional network-based IDS. By employing the HAI dataset as the experimental foundation, the MIDS astoundingly achieved a fault detection accuracy of 99.76% through the judicious utilization of the random forest algorithm. While the MIDS may not proactively thwart network intrusions, it seamlessly complements the Network IDS, thus bolstering the overall security apparatus. The focal point of this inquiry centered on supervised learning methodologies, urging further exploration into the field of unsupervised learning as a potent avenue for unmasking novel anomalies lurking within the system. Khan et al. [28] suggested an approach that transforms network flows into spectrogram images and utilizes CNN for processing. The model achieved impressive detection accuracy (99.35% for binary and 98.76% for multiclass classification) while effectively reducing the False Alarm Rate. Future research aims to address computational complexity, evaluate the approach on diverse datasets, and extend its application to unsupervised learning scenarios for real-world deployment.

Hwang et al. [29] proposed D-PACK, a novel framework for early malicious traffic detection. By leveraging sampling, auto-profiling of the traffic using deep learning models such as Convolutional Neural Networks, and an unsupervised DL model (autoencoder), D-PACK achieved nearly 100% accuracy with minimal false positives and negatives. The framework's key advantage is its ability to speed up detection by examining fewer packets and bytes. The study encourages further optimization methods using deep learning for efficient online anomaly detection systems without significant detection delays. Feng et al. [30] designed a plug-and-play device for detecting XSS, DoS and SQL attacks in ad-hoc networks using a deep learning model. Experimental results showed high detection performance. However, further research is needed to address issues such as collecting more malicious samples and designing an adaptive security architecture (Table 1).

Abdalgawad et al. [31] confronted the escalating apprehensions surrounding cyberattacks targeting Internet of Things (IoT) devices. Their pioneering work shed light on the remarkable efficacy of generative deep learning (DL) methodologies, namely bidirectional generative adversarial networks (BiGAN) and adversarial autoencoders (AAE), in the accurate classification of attacks afflicting IoT devices. To substantiate their claims, the researchers precisely conducted experiments employing the IoT-23 dataset, specially crafted from IoT devices, as the foundation of their investigation. Through a meticulous comparison between conventional baseline models and the formidable GAN-based counterparts, the findings unveiled the undeniable superiority of the GAN models in not only discerning but also proficiently classifying various attacks. Moreover, the researchers embarked on a groundbreaking exploration into the prospect of introducing novel information into the test set, serving as a litmus test to gage the model's prowess in detecting anomalies. Encouragingly, the model showcased commendable success in detecting and pinpointing such atypical scenarios, thus underscoring its robustness and adaptability.

The existing landscape of research on ML based IDSs targeting unknown attacks, particularly in zero-day scenarios, encounters certain limitations that impede progress.

Table 1 Comparative Analysis of Intrusion Detection Systems (IDSs) for Zero-Day Attack Detection

Study	Year	Key Concept	Model	Detection Type	Dataset	Performance Measures	Handles zero-Day
Hwang et al. [29]	2020	Computationally less expensive unsupervised IDS for anomaly detection	CNN, Autoencoder	Behavior-based	USTC-TFC2016, Mirai-RGU, Mirai-CCU	Acc. = 99.77%, F1 = 99.55%	Yes
Feng et al. [30]	2019	IDS for DoS and privacy attacks detection	DNN, CNN, LSTM	Signature-based	KDD CUP99	Acc. = 99.9%, F1 = 0.986	No
Mokhtari et al. [27]	2021	NIDS for Industrial SCADA systems	KNN, RF, DT	Signature-based	Captured live data	Acc. = 99.76%	No
Khan et al. [28]	2021	A spectrogram-based methodology is proposed for the purpose anomaly detection.	CNN	Signature-based	CICIDS2017	Acc. = 98.78%, F1 = 98.67	No
Sabeel et al. [26]	2021	Choosing features in two stages for traffic analysis in a flow-based network	DNN, Linear SVC, Stacked Decision Tree	Behavior-based	CICIDS2017	TPR = 99.90%, TNR = 99.92%	Yes
Abdalgawad et al. [31]	2021	Generative DL-based IDS for cyberattacks detection	AAE, BiGAN, KNN, RF	Behavior-based	IoT-23	Acc. = 97%, F1 = 0.94	Yes
Zhang et al. [22]	2020	Semantic space-based detection of previously unseen or unknown attacks	Sparse Autoencoder	Signature based	NSL-KDD	Acc. = 88.3%	Yes
Li et al. [24]	2019	Zero-day attack detection using random forest and spatial clustering	Random Forest, Spatial Clustering	Hybrid approach	NSL-KDD	Acc.=77.28%	Yes
Kumar et al. [25]	2021	Model for the detection of zero-day attacks using heavy-hitter low-volume attacks and graph technique	Heavy-hitter, Graph Technique	Hybrid approach	Real-time attack dataset	Acc. = 91.62%, F1: 0.95	Yes
HMS-IDS (Proposed)	2023	Model for the detection of zero-day attacks using ensemble learning and K-Means	DT, KNN, LR, GNB & K-Means	Hybrid approach	CIC TON-IOT	Hybrid Acc. & F1 = 98.43%, 0.98 Ensemble Acc. & F1 = 99.49%, 0.98	Yes

These limitations primarily stem from the inadequate level of extensive experimentation conducted on recent datasets encompassing a wide array of attack types. The heavy reliance on outdated datasets, such as the NSL-KDD dataset, engenders legitimate concerns regarding the pertinence and applicability of evaluation outcomes. The utilization of antiquated datasets fails to capture the intricacies of modern network traffic characteristics and the multifaceted nature of contemporary threats, thus undermining the reliability and validity of proposed methodologies. To surmount these obstacles and pave the way for future advancements, it is imperative for forthcoming research endeavors to prioritize the incorporation of up-to-date and diverse datasets that faithfully mirror the prevailing network security landscape. By adopting this strategic approach, the scientific community can conduct more comprehensive evaluations of IDSs, thereby fortifying their detection and mitigation capabilities against sophisticated and ever-evolving cyber threats.

3 Datasets

3.1 CIC-ToN-IoT Dataset

The CIC-ToN-IoT dataset, introduced in [32], serves as a valuable resource for studying IoT network traffic. With the aid of the CICFlowMeter tool, researchers can extract features from pcap files within the ToN-IoT dataset, offering a rich dataset for analyzing and evaluating various methodologies IoT network security field. By leveraging the capabilities of the CICFlowMeter-v4 tool, researchers can extract 83 distinct attributes, providing a comprehensive and detailed characterization of the data. The dataset may be augmented[1] using data augmentation techniques [33] to broaden and diversify the training set, which will further improve the performance metrics. However, the training set of the dataset used in this manuscript is also good enough to train the proposed model, hence the idea of augmenting the training set is proposed as a future scope of this research work (Table 2).

This extensive feature set contributes to a deeper understanding and analysis of network traffic in IoT environments, enabling the development of effective security solutions.

The study utilized the CIC-ToN-IoT dataset, comprising a total of 5,351,760 data samples. Among these samples, 2,836,524 (53.00%) are classified as attacks, while 2,515,236 (47.00%) are categorized as benign instances (Fig. 2). This balanced distribution between attack and non-attack data allows for a comprehensive representation of both classes, facilitating effective analysis and modeling of their respective characteristics and patterns. It is important to emphasize that the dataset was specifically created for the mentioned research study. Researchers interested in accessing and utilizing this dataset can refer to the provided source [32], which offers detailed information on obtaining the dataset for further analysis and exploration.

Table 2 Monitoring the Volume and Distribution of Attack Types in the CIC-ToN-IoT Dataset

Attack Type	Number of Samples
Attack Distribution	
Xss	2,149,308
Password	340,208
Injection	277,696
Scanning	36,205
Backdoor	27,145
Ransomware	5098
MITM	517
DDoS	202
Dos	145

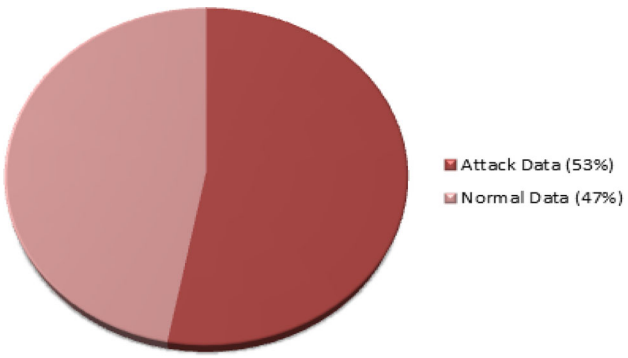


Fig. 2 Unveiling the Balance: Pie Chart depicting Attack vs. Benign Data in the CIC-ToN-IoT Dataset

3.2 KDD-99 Cup Dataset

The KDD-99 cup dataset [34] is widely regarded as the primary dataset for training algorithms. It is a portion of the DARPA-98 dataset. The KDD-99 dataset is a dataset that contains several varieties of data. The dataset comprises 4.8 million occurrences. The attributes utilized in the dataset possess category and integer properties. The object possesses a total of forty-two characteristics. TheKDD-99

- This dataset consists of TCP dump data from simulated network traffic that was gathered in 1998 at Lincoln Labs. After a period of seven weeks, a total of five million connection data were collected and utilized as a training set. An additional two weeks of network traffic resulted in

[1] Data augmentation is a method that uses pre-existing data to create modified copies of a dataset, thereby artificially expanding the training set.

Table 3 Class Distribution in Training and Test Sets

Class	Training Set		Test Set	
	Count	Percentage (%)	Count	Percentage (%)
Normal	812,814	75.611	60,593	19.481
DoS	247,267	23.002	229,853	73.901
Probe	13,860	1.289	4166	1.339
R2L	999	0.093	16,189	5.205
U2R	52	0.005	228	0.073
Total	1,074,992	100	311,029	100

Table 4 NSL-KDD Dataset Distribution

Class	Training Set	Percentage (%)	Test Set	Percentage (%)
Normal	67,342	53.458	9710	43.075
DoS	45,927	36.458	7457	33.080
Probe	11,656	9.253	2421	10.740
R2L	995	0.790	2754	12.217
U2R	52	0.041	200	0.887
Total	125,972	100	22,542	100

the creation of a test set including two million instances. KDD-99 is a refined version of this dataset.

- The KDD-99 dataset consists of five distinct kinds of patterns, namely Normal, DoS (Denial of Service), U2R (User to Root), R2L (Remote to Local), and Probe (Probing Attack). Each intrusion category is further categorized based on the specific method employed to carry out the assault. The assaults are tabulated in Table 3

3.3 NSL-KDD Dataset

In order to address the problems present in the KDD-99 cup dataset, researchers have introduced a new dataset called NSL-KDD. This dataset is composed of carefully chosen records from the original KDD dataset and is free from any of the aforementioned concerns. NSL-KDD is a project initiated by Tavallaee et al. [35] to address the issues of KDD-99 and overcome its limitations. Nevertheless, the authors acknowledge that the dataset still suffers from some issues, including its failure to accurately reflect low footprint assaults (Table 4).

3.4 CIC IDS 2017

The CICIDS2017 dataset [36] accurately replicates network data from real-world scenarios, specifically PCAPs. It employs CICFlowmeter-V3.0 to extract a total of 78 features and 79 labels. The dataset contains the abstract attitudes of 25 users toward the HTTP, HTTPS, FTP, SSH, and email protocols, as displayed in Table 5. Information is collected across various time intervals.

4 Proposed Methodology

4.1 System Architecture

The proposed work (Fig. 3) elucidates a hybrid multi-stage intrusion detection system (HMS-IDS) precisely designed to cater exclusively for the IIoT networks. The structural framework of the system encompasses four discrete stages, namely data pre-processing, feature engineering, a signature-based IDS, and an anomaly-based IDS. In the data pre-processing stage, raw data undergoes cleaning and formatting to ensure its suitability for analysis. Feature engineering involves extracting relevant features from the pre-processed data, allowing the detection system to capture essential characteristics of network traffic. The signature-based IDS utilizes a database of known attack signatures to compare incoming traffic and identify any matches, enabling the detection of familiar or previously documented attacks. Whereas, the second stage unsupervised IDS focuses on identifying abnormal patterns and behaviors in network traffic.

By employing machine learning techniques, it learns the general properties of the network and detects any deviations that may indicate a potential attack, including previously unseen or unknown attack types. Through the integration of these stages, the proposed architecture aims to deliver a comprehensive and robust IDS for IIoT networks. It effectively identifies and mitigates both known and unknown cyber threats, thereby enhancing the overall security of the network environment.

Table 5 Day-wise Labels of CICIDS 2017 Dataset Excluding. The Non Working Days (Saturdays and Sundays)

Days	Labels
Monday	Benign
Tuesday	BForce, SFTP, and SSH
Wednesday	DoS and Heartbleed attacks, Slowloris, Slowhttptest, Hulk, and GoldenEye
Thursday	Web and infiltration attacks, Web BForce, XSS, and Sql Inject. Infiltration Dropbox Download and Cool disk
Friday	DDoS LOIT, Botnet ARES, PortScans (sS, sT, sF, sX, sN, sP, sV, sU, sO, sA, sW, sR, sL, and B)

4.2 Data Pre-processing

The proposed system encompasses an initial data pre-processing stage, aimed at precisely cleansing and standardizing the raw network data. Subsequently, in the feature engineering stage, a comprehensive set of 33 distinct features is extracted, encompassing a wide range of statistical attributes pertaining to flows, protocols, and packet lengths. These features facilitate efficient identification and examination of prospective vulnerabilities in IIoT networks. The pre-processing of raw data and developing of custom features through the feature engineering process, improve the accuracy of intrusion detection and network security by finding both known and unknown threats.

4.2.1 Data Sampling

The significance of data sampling in enhancing the efficacy of ML model training on vast volumes of network traffic data is paramount. By creating a reduced subset from the original dataset, sampling methodologies alleviate the computational burden entailed in handling extensive data, thereby facilitating expeditious and efficient model training. Within the proposed system, a cluster sampling approach, predicated on the principles of k-means clustering, is implemented to procure a representative subset of data. This technique ensures that the selected subset adeptly encapsulates the diversity and intrinsic attributes exhibited by the original dataset. During the cluster sampling process, data points are methodically clustered into distinctive groups, from which an appropriately proportionate sample is extracted, thereby culminating in the formation of the representative subset. This approach permits the selection of varied data samples that encompass a myriad of characteristics observed throughout the entirety of the dataset. By harnessing the amalgamation of cluster sampling and k-means clustering, the proposed system endeavors to streamline the training process while optimizing the utilization of network traffic data for the development of ML models.

In contrast to random sampling, where each data sample is selected with equal probability, cluster sampling offers the advantage of generating a highly representative subset by effectively discarding redundant data. This technique leverages the clustering of data points to identify groups or clusters that share similar characteristics. K-means is widely regarded as the most commonly used clustering algorithm for data sampling.

K-means clustering algorithms utilize distance metrics such as Euclidean, Manhattan, or Mahalanobis distances to partition data points into k clusters. By measuring the dissimilarity between data points, the algorithm identifies the closest centroid for each point, assigning it to the corresponding cluster. This process is repeated iteratively, refining the cluster assignments until convergence is reached. This approach acknowledges that samples within a group share common characteristics or patterns, enabling the selection of representative samples that encapsulate the essential information present in the complete dataset.

The K-means algorithm endeavors to minimize the aggregate of squared distances linking individual data points to their respective assigned cluster centroids. Through a series of iterative steps, it dynamically adjusts the centroids to attain the optimal configuration that aptly captures the inherent structure exhibited within the data. The formula for calculating this sum of squared distances is:

$$\sum_{i=0}^{n_k} \min_{u_j \in C_k} \left(\mathbf{x}_i - u_j\right)^2,$$

The K-means algorithm operates on a data matrix $(x_1, \ldots, x_n)$, where each x_i represents a data point. In this algorithm, u_j, also called the cluster's (C_k) "centroid",

is calculated as a sample's weighted average (mean) in C_k. Additionally, n_k sums up the number of data points used to form the cluster. C_k. K-means algorithm's time complexity is linear and can be represented as $O(nkt)$, where n refers to

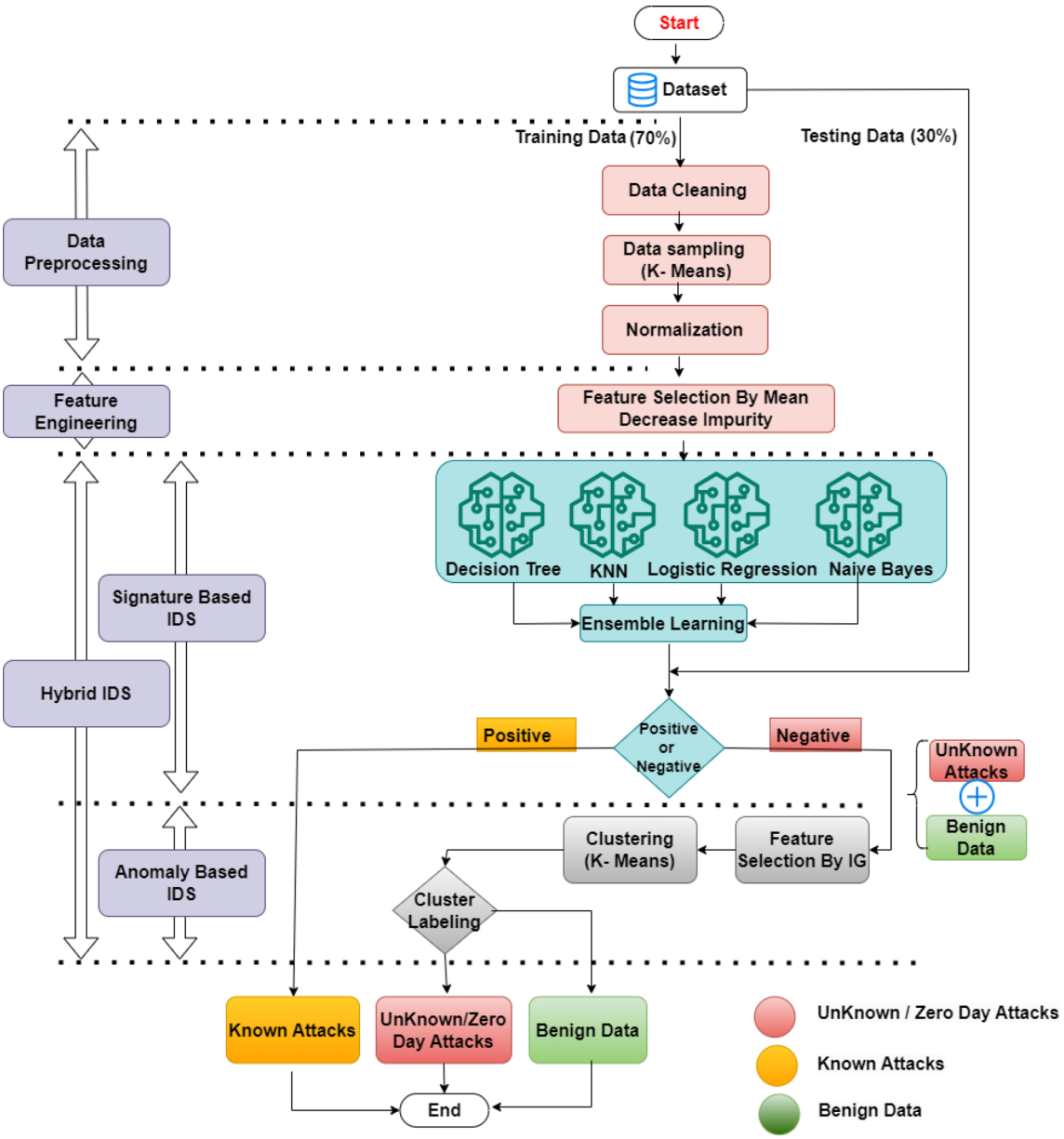


Fig. 3 Proposed Methodology Diagram of the HMS-IDS Contains Phase I. where 70% of the Dataset is used as Training Dataset and supplied for DATA Preprocessing, Phase II. Involves Feature Engineering Part and at Last the Ensemble Model is Trained and Tested to Detect the Known Attacks also called as Signature Based IDS and Remaining Undetected Portion is used to Send to Unsupervised Model. After the Feature Selection Process, the Final Categorization is Done and the Benign and Anomaly/Unknown Attacks/ZDAs are Detected

the size of the data, k represents the number of clusters, and t denotes the number of iterations.

4.2.2 Data Normalization

Upon the completion of the representative dataset acquisition through the utilization of the k-means method, a subsequent series of data pre-processing procedures is initiated to prepare for the ensuing stages. These ensuing steps encompass a range of data manipulation and transformation techniques aimed at enhancing the quality and relevance of the dataset. As an initial step, the network traffic datasets are subjected to encoding utilizing a label encoder, thereby facilitating the conversion of categorical attributes into numerical representations, thereby enabling the subsequent employment of machine learning algorithms. This process is deemed imperative due to the inherent requirement of numerous machine learning algorithms for numeric inputs, rendering them incapable of directly processing string or categorical attributes. Subsequently, subsequent to the label encoding

procedure, the network datasets undergo normalization utilizing the Z-score algorithm, a method indispensable for achieving standardized and comparable data by adjusting each attribute's values relative to the dataset's mean and standard deviation.

This normalization technique is employed due to the significant variation in range among the collected features in network traffic data. Normalizing the data to a common scale is beneficial for ML models as it can enhance their performance. When working with an unnormalized dataset that exhibits significant differences in feature scales, there is a risk of introducing bias into the machine learning model. This bias occurs because the model tends to prioritize features with larger scales over those with smaller scales, potentially overshadowing the impact of the latter. Consequently, the model's decision-making process may become skewed, relying heavily on the dominant features while undervaluing the contributions of others. By applying the Z-score method, the features within the dataset are transformed to possess a mean value of 0 and a standard deviation of 1. This normalization process standardizes the feature values, aligning them with a common scale.

By incorporating the Z-score technique, the normalization of each feature value, represented as x_n, is expressed as follows:

$$x_n = \frac{x - \mu}{\sigma}$$

In the feature normalization process, each original feature value, denoted as x, is transformed using the mean (μ) and standard deviation (σ) of the feature values. This transformation ensures that the features are on a similar scale and have a comparable range, which aids in the accurate analysis and modeling of the data.

4.3 Feature Engineering

After undergoing a data pre-processing phase, a dataset of exceptional quality and strong representativeness is generated. Furthermore, by employing effective feature engineering techniques [37], it is possible to attain an optimal list of features that greatly enhance the quality of the datasets. Feature engineering involves selecting, transforming, or creating new features based on domain knowledge and data analysis. Before training, the machine learning model, a comprehensive feature engineering method, including information gain (IG), is executed to streamline the dataset. This method serves the purpose of facilitating the process of feature selection by effectively discerning and eliminating inconsequential, duplicative, and disruptive attributes, while simultaneously preserving the pivotal ones that exert substantial influence on the desired outcomes [38].

The method of feature Selection by information gain(IG) is extensively utilized as a prevalent technique for the process of feature selection (FS). By utilizing the IG method, the objective is to ascertain and select the most salient features from the given dataset. The notion of IG centers on the quantification of the degree to which a characteristic contributes valuable information or diminishes uncertainty with respect to the target variable. Through the assessment of entropy changes, which quantifies the level of randomness or unpredictability, the IG (Information Gain) method evaluates the informational contribution of each feature [38]. The choice of the IG method in the proposed system is predicated on its advantageous characteristics, such as its capacity to expeditiously compute importance scores for each feature. The IG (Information Gain) method is renowned for its minimal computational complexity, represented as O(n), thereby facilitating efficient processing, particularly for extensive datasets [38]. By applying a scoring mechanism to each feature, we acquire the capability to ascertain the significance of features about the specified task. These importance scores serve as a valuable metric for evaluating the contribution of each feature to the desired outcome.

Assuming T represents the target variable, the IG of utilizing the random variable X as a feature is denoted by [9]. The IG value quantifies the reduction in uncertainty about the target variable T achieved by incorporating the feature X in the analysis. It measures the extent to which X provides new information or discriminative power for predicting T.

$$T(T \mid X) = \Psi(T) - \pi(T \mid X)$$

The IG value of the target variable T over a feature X, denoted as IG($T|X$), is defined as the difference between the entropy of the target variable T ($H(T)$) and the conditional entropy of T given X ($H(T|X)$) [39]. It quantifies the importance of feature X to the target variable T, indicating the reduction in uncertainty about T achieved by incorporating feature X in the analysis. In particular, if IG($T|X$) > IG($T|Y$), where Y represents another feature, it signifies that feature X is given higher importance to the target variable T than feature Y [40].

4.4 Proposed Hybrid Intrusion Detection System Design

Intrusion detection systems have two prominent classifications, namely signature/Rule-based and Behavior/Anomaly/Heuristics-based. Rule-based IDS is precisely crafted to discern and acknowledge well-established attack patterns through the employment of machine learning models trained on datasets annotated with labels that correspond to preconceived attack patterns. Although these systems exhibit commendable effectiveness in detecting familiar attacks,

they often confront impediments when it comes to discerning uncharted and burgeoning attack patterns that have yet to be cataloged or archived within their databases [41].

In contrast, Anomaly Heuristics-based IDSs leverage unsupervised learning algorithms to differentiate unknown attack data from normal data. These systems function under the assumption that recently encountered attack data will demonstrate statistical resemblances with known attack data, as opposed to normal data. Nevertheless, a prevalent obstacle encountered by anomaly-based IDSs is their inclination to produce a substantial quantity of false alarms [42]. In order to address the constraints of standalone IDSs, this research presents a groundbreaking hybrid IDS that combines both Heuristics and Rule-based methodologies. Through the amalgamation of the inherent capabilities of these two methodologies, the hybrid IDS endeavors to furnish an all-encompassing and efficacious detection mechanism, catering to both pre-existing attacks and novel zero-day attacks.

4.4.1 The Signature-Based IDS (To Identify Known Attacks)

Once the steps for feature engineering and data pre-processing are finished, the labeled datasets are utilized to train an ensemble learning model, which serves as the fundamental component of the signature-based IDS. By utilizing ensemble learning, which amalgamates multiple individual classifiers, the IDS is capable of augmenting its detection capabilities and enhancing overall accuracy. The signature-based IDS under consideration integrates a heterogeneous ensemble of machine learning algorithms as its foundational learners. Four ML algorithms, particularly Naive Bayes (NB), decision tree (DT), Logistic Regression (LR), and k-Nearest Neighbors (KNN) have been chosen based on their unique characteristics and advantages [27]. Every individual base learner undergoes training using labeled datasets, enabling them to effectively capture distinct characteristics of the underlying data and make unique contributions to the as a whole process of identification.

(i) **Logistic Regression (LR)**: Binary categorization uses Logistic regression. The Logistic function estimates the binary outcome probability based on input characteristics. The Logistic function, also known as the sigmoid function, is a key component of Logistic regression and converts every real-valued input to a probability of the positive class between 0 and 1. The sigmoid function is defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where z is a linear combination of the input features and their corresponding weights:

$$z = w_0 + w_1 x_1 + w_2 x_2 + \ldots + w_n x_n$$

Here:

- w_0 is the bias term,
- $w_1, w_2, \ldots, w_n$ are the weights associated with the input features $x_1, x_2, \ldots, x_n$.

(ii) **Naive Bayes (NB)**: NB is an algorithm for classification that operates based on the principles of probability theory, particularly Bayes' theorem. The features in the dataset are assumed to be conditionally independent, which means that the presence or absence of one feature has no bearing on the probability that another feature would be present or absent. The Naive Bayes classifier computes the probability of a class C_k given the input features using Bayes' theorem:

$$P(C_k|X) = \frac{P(X|C_k) \cdot P(C_k)}{P(X)}$$

Here:

- $P(C_k|X)$ is the posterior probability of class C_k given the features X.
- $P(X|C_k)$ is the likelihood of observing the features X given class C_k.
- $P(C_k)$ is the prior probability of class C_k.
- $P(X)$ is the evidence, the probability of observing the features X.

The "naive" assumption in Naive Bayes is that features are conditionally independent given the class label. Mathematically, this can be expressed as:

$$P(X|C_k) = P(x_1|C_k) \cdot P(x_2|C_k) \cdot \ldots \cdot P(x_n|C_k)$$

Under this assumption, the Naive Bayes classifier simplifies to:

$$P(C_k|X) = \frac{P(C_k) \cdot \prod_{i=1}^{n} P(x_i|C_k)}{P(X)}$$

The class with the highest posterior probability is then predicted as the output.

(iii) **Decision Tree (DT)**: The DT algorithm, being a commonly utilized technique in machine learning, utilizes a hierarchical structure resembling a tree to scrutinize and categorize data. By employing recursive partitioning

techniques on the input data, the algorithm generates a collection of decision rules that can be utilized for prediction purposes [43]. When employing DT algorithms, it is imperative to carefully adjust multiple hyperparameters to achieve optimal performance. Some examples of hyperparameters include the maximum number of leaf nodes, the minimum fraction of total sample weights needed for a leaf node, the minimum number of samples needed to divide an internal node, and the minimum number of samples needed to form a leaf node [43].

(iv) **K-Nearest Neighbors (KNN)**: In ML, the KNN method is a frequent choice for classification problems because to its nonparametric nature.It assigns a new data point to the feature space class with the most kNNs. For classification tasks, let x be a new data point, and $N_k(x)$ represent the set of k-nearest neighbors of x in the training set. The predicted class y for x can be obtained by a majority vote:

$$y = \text{argmax}_{c_i} \sum_{x_i \in N_k(x)} I(y_i = c_i)$$

Here:

- c_i is a class label,
- y_i is the class label of the i-th neighbor,
- $I(\cdot)$ is the indicator function that returns 1 if the condition is true and 0 otherwise.

For regression tasks, the predicted value is often the average (mean) of the values of the k-nearest neighbors:

$$\hat{y} = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i$$

Assuming n instances, f features, and t base learners in ensemble models, the time complexity of DT, NB, KNN, and Logistic regression is t, the time complexity of DT, NB, KNN, and Logistic regression is $O(n2f)$, $O(nf)$, $O(nft)$, and $O(nf)$, respectively [44]. Factors behind the selecion of algorithms are as below:

1. DT, NB, KNN, and Logistic regression algorithms frequently exhibit superior performance compared to tree-based algorithms when dealing with uncomplicated datasets.
2. They have relatively lower time complexity than tree-based algorithms, especially for NB and Logistic regression.
3. They have fewer hyper-parameters to tune, making it easier to find a good model.
4. Using different types of base learners can lead to a more diverse ensemble model, which often results in better performance.

Algorithm 1: Signature-Based IDS

```
function SignatureBasedIDS(train_data)
    // Feature engineering and data pre-processing
    processed_data ← PreprocessData(train_data)
    // Ensemble learning with base learners
    base_model_DT ← TrainDecisionTreeClassifier(processed_data)
    base_model_NB ← TrainNaiveBayesClassifier(processed_data)
    base_model_KNN ← TrainKNearestNeighborsClassifier(processed_data)
    base_model_LR ← TrainLogisticRegressionClassifier(processed_data)
    // Stacking ensemble to combine base models
    stacked_model ← TrainStackingClassifier(base_model_DT, base_model_NB,
    base_model_KNN, base_model_LR, processed_data)
    return stacked_model
```

- **Ensemble Model:** To enhance the overall model performance, the four base models are integrated using stacking, an ensemble learning technique. Stacking combines the predictions of multiple base learners, leveraging their individual strengths and generalizability [45]. To leverage the predictions of the four base learners (DT, NB, KNN, and Logistic regression), the stacking ensemble learning technique is employed. Stacking involves utilizing the estimated output labels from these base learners as input features for training a meta-learner [45]. By employing the stacking technique, the proposed system aims to leverage the collective knowledge from all four base learners to minimize individual errors and enhance the overall reliability and robustness of the meta-classifier. In this approach, the best-performing base model among the four is selected as the algorithm for constructing the meta-learner.

4.4.2 The Anomaly-Based IDS (To Cluster Unknown Attacks and Benign Data)

The signature-based IDS introduced in this study demonstrates effective detection capabilities for various types of known attacks. Despite the effectiveness of the proposed signature-based IDS in detecting known attacks, there remains a vulnerability to zero-day attacks, which are not captured by the existing known attack patterns and may be mistakenly classified as normal system behavior.

As a result, instances labeled as "normal" by the signature-based IDS are regarded as potentially suspicious, as they may include unknown attack samples that were not captured by the known attack patterns. Subsequently, a novel architecture for an anomaly-based IDS is devised, focusing on the analysis and processing of the potentially suspicious instances to detect zero-day attacks.

Once the feature engineering stage is completed, the optimized dataset resulting from the feature selection process using the IG method is utilized for training the anomaly-based IDS. The initial tier of the proposed anomaly-based IDS involves the utilization of an enhanced version of the k-means algorithm known as cluster labeling (CL) k-means. Given the extensive range of instances present in network traffic datasets, it is crucial to employ an adequate number of clusters to effectively differentiate between benign and attack data.

The following are the major steps in the suggested CL-k-means method:

1. *K-Means Clustering*: It divides the dataset into an ample number of clusters is achieved through the utilization of the k-means algorithm.

$$\text{argmin}_{C_1,C_2,\ldots,C_k} \sum_{i=1}^{k} \sum_{x \in C_i} \|x - \mu_i\|^2$$

where μ_i is the centroid of cluster C_i.
2. *Labeling Clusters*: The clusters are labeled based on the majority class label of the data samples they contain. Each cluster is assigned the class label ("normal" or "attack") that represents the majority of instances within that cluster.

$$M_i = \text{argmax}_{\text{class}} \sum_{x \in C_i} I(\text{label}(x) = \text{class})$$

where $I(\cdot)$ is the indicator function.
3. *Assigning Labels to Test Samples*: Each sample in the test set is assigned a label ("normal" or "attack") based on the label of the cluster to which it belongs, determined by the k-means algorithm

$$\text{label}(x_i) = M_j \quad \text{where} \quad x_i \in C_j$$

4. *Computing Confidence or Clustering Probability*: To assess the confidence or clustering probability of each test sample i within its assigned cluster, the percentage of majority class samples in that cluster is computed.

$$\text{Confidence}(x_i) = \frac{\text{Number of majority class samples in } C_j}{\text{Total number of samples in } C_j} \times 100$$

The mentioned approach leverages the k-means algorithm for real-time demands in distinguishing between attack and normal data in IIoT-level systems. The efficiency of k-means, with a linear time complexity of $O(nkt)$, is a key factor in this choice, where n is the size of the data, k is the number of clusters, and t is the number of iterations [46].

Algorithm 2: Anomaly-Based IDS

```
function AnomalyBasedIDS(train_data)
    // Feature engineering and data pre-processing with Information Gain
    processed_data ← PreprocessDataWithInformationGain(train_data)
    // CL-k-means clustering
    clusters ← KMeansClustering(processed_data)
    // Labeling clusters based on majority class
    labeled_clusters ← LabelClusters(clusters)
    // Assigning labels to test samples based on cluster membership
    test_labels ← AssignTestLabels(processed_data, labeled_clusters)
    // Calculate clustering probability for each test sample
    clustering_probabilities ← ComputeClusteringProbabilities(processed_data, labeled_clusters)
    return test_labels, clustering_probabilities
```

To further decrease the training time of the model, minibatch k-means is employed, which employs randomly selected subsets, known as minibatches, during each iteration of training [47]. Furthermore, k-means provides the advantage of convergence assurance and seamless adaptation to new samples.

The proposed CL-k-means technique with biased classifiers has numerous benefits in terms of obtaining high accuracy and efficiency compared to other unsupervised anomaly detection algorithms like one-class SVM (OCSVM) [48] and iso-lation forest (iForest).

1. The CL-k-means model, as proposed, demonstrates its ability to utilize a large number of clusters, allowing for effective capturing of various attack and normal patterns present in the data samples. This unique feature enhances the model's generalizability and improves its capability to accurately model data patterns compared to binary models employed in other outlier detection techniques like iForest and OC-SVM.
2. The CL-k-means model can integrate the BO-GP optimization technique to automatically estimate the ideal number of clusters (k) required for effective data analysis. This adaptive strategy empowers the proposed model to adapt to diverse datasets and tasks by dynamically adjusting the complexity of the learned data patterns. Lever-

Algorithm 3: K-Means-based Data Classification for IIoT-level Systems

```
procedure KMeansBasedClassification(D, k, t)
    // Step 1: Apply k-means algorithm
    centroids ← InitializeRandomCentroids(D, k) for iteration = 1 to t do
    end
    clusters ← AssignSamplesToClusters(D, centroids)
    centroids ← UpdateCentroids(clusters)

    // Step 2: Label the clusters
    cluster_labels ← [] for each cluster in clusters do
    end
    cluster_label ← MajorityClassLabel(cluster)
    cluster_labels.append(cluster_label)

    // Step 3: Assign labels to test samples
    labeled_samples ← [] for each sample in test_set do
    end
    nearest_cluster ← FindNearestCluster(sample, centroids)
    label ← cluster_labels[nearest_cluster]
    labeled_samples.append(label)

    // Step 4: Compute clustering probability (optional)
    clustering_probabilities ← ComputeClusteringProbabilities(test_set, centroids, clusters)
    return labeled_samples
```

aging this adaptive mechanism enhances the model's performance and flexibility in accurately capturing the inherent structures within the data.

3. By employing the mini-batch k-means algorithm, the IDS effectively accomplishes noteworthy decreases in execution time, thereby fulfilling the demanding real-time necessities of IIoT environments. This optimization technique enables the model to perform data processing in smaller subsets, referred to as mini-batches, within each iteration. As a result, it reduces the computational workload and improves the efficiency of the system. The IDS is capable of expeditiously analyzing and reacting to network traffic, thereby guaranteeing the timely and efficient identification of potential intrusions in IIoT environments.

The presence of attack patterns that closely resemble normal patterns poses a challenge in accurately distinguishing between them. Furthermore, during specific legitimate network events such as crowd events, the collected samples may exhibit significantly different patterns compared to existing normal patterns, leading to potential misclassification as attack samples. Moreover, while k-means is often preferred for developing IDSs in the context of IIoT due to its computational efficiency, it is important to acknowledge the inherent limitation of k-means in accurately modeling complex data distributions. This limitation should be carefully addressed for improved performance and reliability [48].

Given sufficient time and resources, it is possible to explore alternative clustering algorithms that are better suited for specific data shapes and distributions. By replacing k-means with such algorithms while retaining the cluster labeling technique,the system's functionality as a whole has room for improvement. We plan to investigate the potential of online learning techniques to continually update the learning models based on emerging attack patterns. By incorporating this capability into the IDS, it is expected to enhance its generalizability and accuracy over time, as it will be able to adapt and respond to new and evolving threats more effectively.

4.5 Runtime Complexity

During the training phase of the Hybrid IDS, a high-performance server machine is utilized due to its computational capabilities and ample resources. This facilitates the optimization of training the IDS models by leveraging massive data sets and sophisticated algorithms. In the context of the testing phase, it is necessary to deploy and execute the IDS within the operational IIoT systems. This deployment facilitates the capability to monitor and detect network traffic in real-time. This methodology guarantees the seamless integration of the IDS within the IIoT ecosystem, facilitating prompt detection and response to potential security breaches. In order to guarantee smooth functionality within IIoT systems, the suggested IDS prioritizes the creation of models that possess minimal runtime complexity. By optimizing the computational time and reducing the latency of the IDS, the objective of the system is to attain efficient and prompt identification of attacks, while minimizing the impact on the performance of IIoT devices and networks. In the implementation phase of the Hybrid Intrusion Detection System (HIDS), a stacked model is employed to process every test sample. This model combines four supervised ML algorithms, namely DT, KNN, LR, and NB. The utilization of this stacked model, in conjunction with the CL-k-means algorithm, facilitates efficient examination and categorization of network traffic. The runtime Complexity of the Signature-based IDS, CL-k-means Algorithm and Overall Runtime Complexity of the Proposed IDS are given below:

1. *Runtime Complexity of Signature-based IDS (O(nlog(n)f)):* The upper bound on the runtime complexity of the signature- based IDS, which incorporates four supervised machine learning algorithms (decision trees, k-nearest neighbors, Logistic regression, and Naive Bayes), can be expressed as $O(nlog(n)f)$.
   Here:

- n is the number of tuples in the dataset
- f denotes the number of features in the dataset.

2. *Runtime Complexity of CL-k-Means Algorithm (O(ntk))*: The CL-k-means algorithm, incorporated in the proposed IDS, has a runtime complexity of O(ntk). Here:
   - n is the number of tuples in the dataset
   - t is the number of iterations.
   - k represents the number of clusters used in the algorithm.
3. *Overall Runtime Complexity of the Proposed IDS (O(n log(n)f + ntk))*: The overall runtime complexity of the proposed IDS is relatively low, specifically with a maximum of $O(nlog(n)f + ntk)$. This takes into account both the signature based IDS and the CL-k-means algorithm.

The proposed IDS prioritizes creating models with minimal runtime complexity to optimize computational time and reduce latency. The overall efficiency and suitability of the IDS for real-time applications are evaluated, considering computation and categorization time for data samples.

4.6 Evaluation Metrics

For the Model's performance, it is imperative to acknowledge that a singular metric cannot comprehensively gage the excellence of a specific model. Therefore, multiple metrics [15] are employed to assess and analyze the performance of the models.

In the context of our research, a set of metrics has been employed to assess the performance of the proposed predictive model. True positives (TP) represent instances where our model correctly identifies and predicts attack samples within the dataset. Conversely, false positives (FP) indicate cases where the proposed model incorrectly predicts an attack when the sample is, in fact, normal. True negatives (TN) signify the accurate prediction of normal samples, while false negatives (FN) are instances where the model wrongly predicts a sample as normal when it is, in reality, an attack. These metrics are instrumental in evaluating the effectiveness and reliability of our predictive algorithms within the scope of this research work.

(1) *Accuracy* Ratio of correctly predicted samples to total samples.

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

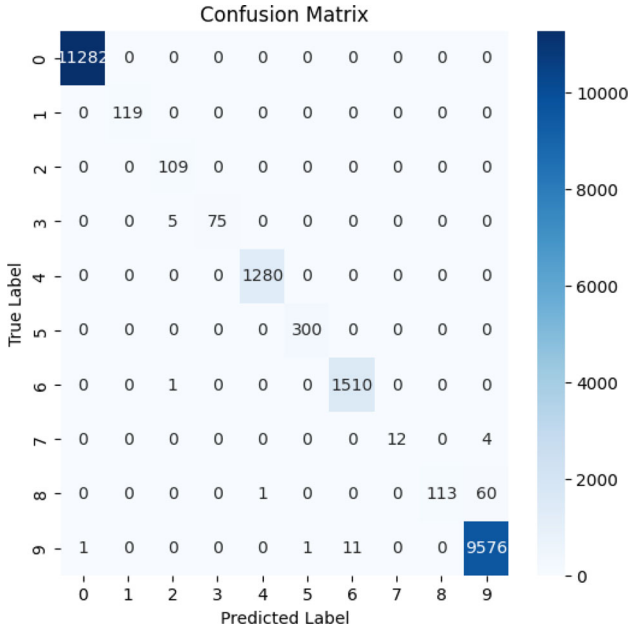


Fig. 4 Confusion Matrix for Signature-Based IDS

(2) *Precision* Ratio of correctly predicted attack samples to total predicted attack samples.

$$\text{PR}\frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

(3) *Recall* Ratio of correctly predicted attack samples to total number of attack samples

$$\text{RE} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

(4) *F1 score* Weighted mean of precision and recall.

$$\text{F1 Score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \tag{4}$$

(5) *Weighted avg* The data points which have higher frequency contribute more than others.

5 Results and Observations

To assess the efficacy of the proposed IDS for accurately identifying known attacks, the ML models incorporated within the signature-based IDS undergo with an extensive training and evaluation procedure utilizing a 10fold cross-validation approach on the CIC-ToN-IoT dataset. This rigorous approach ensures a robust and comprehensive assessment of the models' intrusion detection capabilities by partitioning the dataset into ten equally sized subsets. Subsequently, the

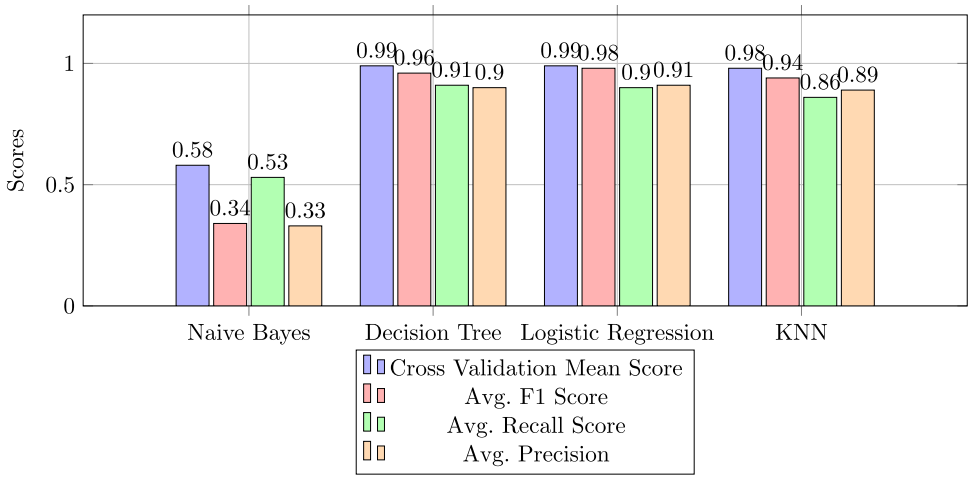


Fig. 5 Signature-Based IDS's Training Accuracy

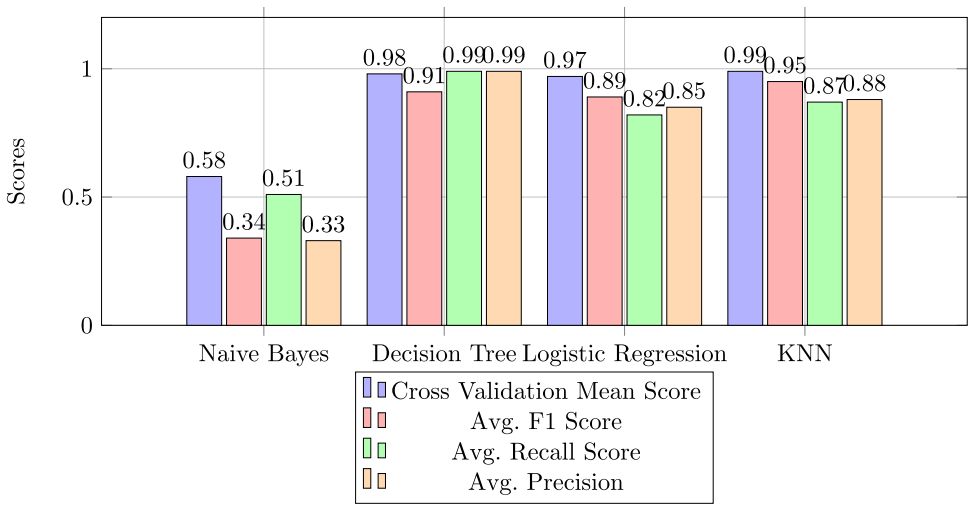


Fig. 6 Signature-Based IDS's Testing Accuracies

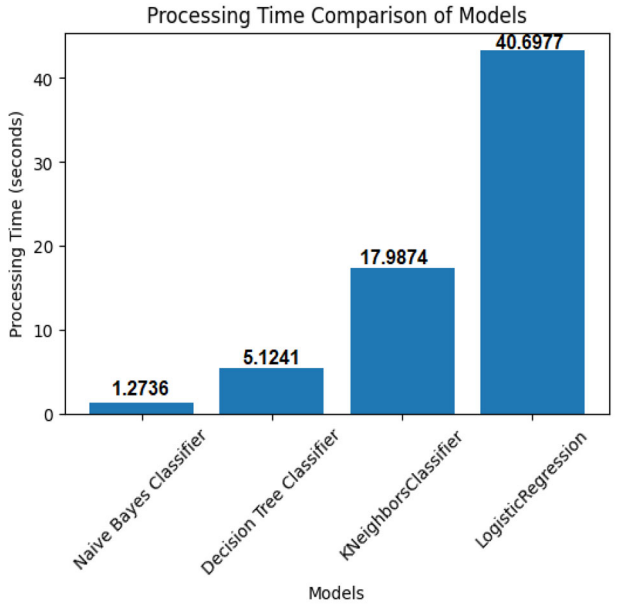
models are trained on nine subsets while their performances are precisely evaluated on the remaining subsets. Through the repetition of this process for all possible combinations of subsets, the evaluation yields reliable metrics regarding the IDS's proficiency in detecting known intrusions across diverse sets of data samples.

5.1 Performance Evaluation of All Individual ML Models on CIC-ToN-IoT Dataset

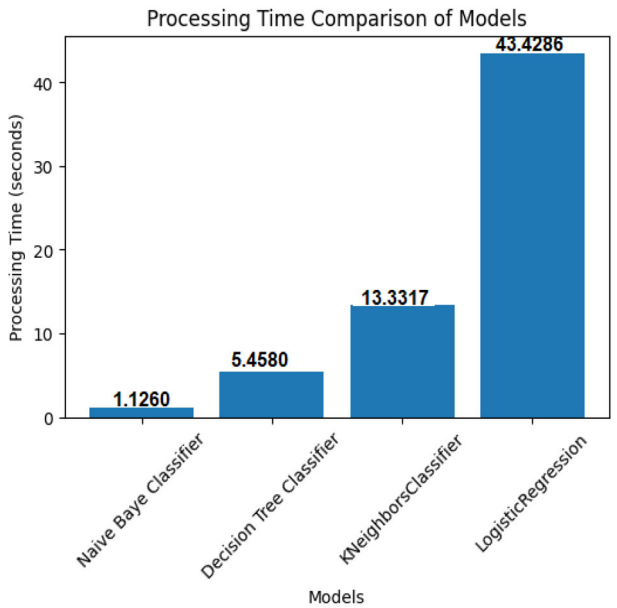
The confusion matrix depicted in Fig. 4 for signature-based IDS is an essential tool for assessing a classification model's strengths and weaknesses since it enables you to fine-tune it, determine the right threshold, or decide between other models depending on your unique objectives and needs and gives a clear image of how well the model performs when separating different classes or categories. The training and testing performance evaluation of the individual ML models like K-Nearest Neighbors, Decision Tree, Random Forest, Naive Bayes is also shown in Figs. 5 and 6 with the respective training and testing time metrics (Fig. 7).

5.2 Performance Evaluation of Ensemble Learning Models on CIC-ToN-IoT Dataset

Based on the tabulated results in Table 6, it is visible that the ensemble approach has not only increased the detection accuracy significantly (99.49%) but also the other performance metrics like F1-score, Recall score, precision, training time, and testing time. In conjunction with assessing the IDS's effectiveness in detecting diverse attack types (Tables 7 and 8), this model is specifically tailored to discriminate between normal and abnormal network traffic data, assigning a categorical label of either "normal" or "abnormal" to each instance. Notably, the proposed IDS attains a remarkable accuracy rate of 99% during the training phase of the binary classification model (Fig. 5) and performed the same during the testing phase (Fig. 6). This aspect affords users the flexibility to opt for either binary classifiers or multi-classifiers, based on their unique prerequisites and preferences.



(a) Training Time (Model-Wise)



(b) Testing Time (Model-Wise)

Fig. 7 Signature-Based IDS's Training and Testing Time-Model-Wise

Table 7 Ensemble Model's Training Multi-Classification Results

Attack	Precision	Recall	F1-Score	Support
Benign	1.00	1.00	1.00	11,241
Xss	1.00	1.00	1.00	120
Password	0.96	1.00	0.98	111
Injection	0.95	0.97	0.96	87
Scanning	1.00	1.00	1.00	1255
Backdoor	1.00	0.99	0.99	285
Ransomware	0.99	1.00	0.99	1566
MITM	1.00	0.86	0.93	22
DDoS	0.97	0.54	0.69	173
DoS	0.99	1.00	0.99	9600
Accuracy			1.00	24,460
macro avg	0.99	0.93	0.95	24,460
weighted avg	1.00	1.00	0.99	24,460

5.3 Performance Evaluation of K-Means Clustering on CIC-ToN-IoT Dataset

The confusion matrix depicted in Fig. 8 for anomaly-based IDS, helps in assessing the clustering model's strengths and weaknesses since it enables you to fine-tune it, determine the right threshold, or decide between other models depending on your unique objectives and needs and gives a clear image of how well the the model performs when separating different classes or categories Based on the tabulated results in Table 9, the implementation of the proposed IG feature selection method and mean decrease impurity method, utilizing the optimized correlation threshold ($\alpha = 0.49$), led to the identification and selection of 15 features from the original set of 85 features. Consequently, the F1-score of the IDS, as applied to the CIC-ToN-IoT dataset, exhibited a noteworthy improvement, reaching a value to 0.99 and the detection accuracy for Zero-day attacks to 99.49%. These findings underscore the crucial role played by the selected features in augmenting the accuracy and efficacy of the IDS, as evidenced by the F1-score metric, thereby enabling the system to accurately detect and classify intrusions.

5.4 Model Validation on Different Datasets

For the validation of the proposed model, the performance evaluation of the HMS-IDS model has been also performed.

Table 6 Ensemble-Based Model's Accuracy

Test-Model	Cross Validation Mean Score	Avg. F1 Score	Avg. Recall Score	Avg. Precision	Avg. Accuracy (%)	Training Time (s)	Testing Time (s)
Ensemble	0.99	0.98	0.92	0.94	99.49	16.786	8.885

Table 8 Ensemble Model's Testing Multi-Classification Results

Class	Precision	Recall	F1-Score	Support
Benign	1.00	1.00	1.00	11,241
Xss	1.00	1.00	1.00	120
Password	0.70	1.00	0.83	111
Injection	0.89	0.50	0.64	87
Scanning	1.00	1.00	1.00	1255
Backdoor	1.00	0.97	0.99	285
Ransomware	0.99	1.00	0.99	1566
MITM	1.00	0.88	0.93	22
DDoS	1.00	0.56	0.72	173
DoS	0.99	1.00	0.99	9600
Accuracy			0.9949	24,460
Macro avg	0.96	0.89	0.91	24,460
Weighted avg	0.99	0.99	0.99	24,460

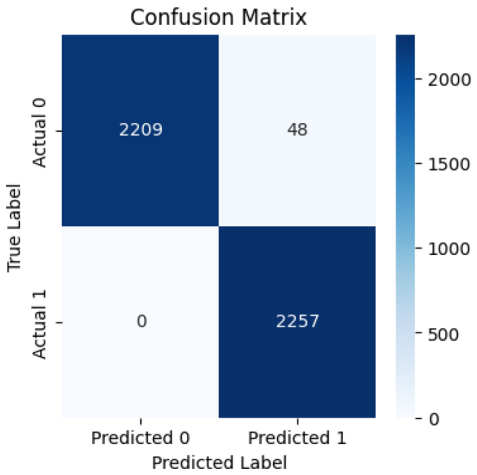


Fig. 8 Confusion Matrix for Anomaly-Based IDS

Table 9 Anomaly Based Model's Detection Accuracy

Class	Accuracy	Precision	Recall	F1-Score	Support
Benign	98.93	1.00	0.98	0.99	2257
Attack	98.93	0.98	0.99	0.99	2257

these datasets are namely KDD-99 Cup, NSL-KDD and CICIDS 2017.

5.5 Performance Evaluation of Signature Based on Different Datasets

Table 10 provides a comprehensive performance evaluation of a Signature-Based Intrusion Detection System (IDS) across various datasets. The evaluation encompasses different machine learning models, namely Naive Bayes, Decision Tree, KNeighbors, Logistic Regression, and an Ensemble model. The assessment is conducted based on key metrics such as Cross-Validation (CV) Mean Score, Accuracy, Precision, Recall, and F1-Score.

5.6 Performance Evaluation of Anomaly-Based IDS Metrics on Different Datasets

Table[11] outlines the performance metrics of Anomaly-Based Intrusion Detection Systems (IDS) across various datasets. The metrics include Accuracy, Precision, Recall, F1-Score, True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN).

5.7 Performance Evaluation of Testing Time Comparison for Signature and Anomaly-Based IDS on Different Datasets

Table[12] presents a comparative analysis of the performance evaluation for Testing Time in Signature-based and Anomaly-based Intrusion Detection Systems (IDS) across various datasets. The metrics considered include the time taken for Support Cluster Creation, Classification time, and the overall testing Time for both types of IDS. This comprehensive evaluation provides insights into the efficiency of both Signature-based and Anomaly-based IDS across diverse datasets. The comparison includes the time taken for critical processes such as Support Cluster Creation and Classification, contributing to a holistic understanding of the testing time performance in different scenarios.

6 Conclusion

The principle idea behind the Hybrid IDS proposed for its application in IIoT networks involves multiple tiers, each serving a distinct purpose to enhance detection capabilities for both known and unknown attacks. The first tier is a signature-based IDS that uses rule-based detection mechanisms to identify predefined attack patterns documented in databases. The second tier is an anomaly-based IDS that employs unsupervised learning techniques to identify deviations from normal behavior, enabling the detection of unknown / Zero Day attacks. The proposed hybrid IDS has been comprehensively evaluated on a real-world dataset and has demonstrated efficacy and robustness in detecting various types of attacks targeting IIoT networks. The third tier of the hybrid IDS, which includes the CL-k- means unsupervised model, is specifically designed to address the detection of unknown or zero-day attacks. It uses the k-means clustering algorithm to group similar network traffic patterns and identify deviations from established norms. The proposed IDS uses a multi-tiered approach that includes the CL-k-means model. It has a high accuracy rate and F1-score,

Table 10 Results of signature-based IDS on different datasets

Dataset	Model	CV Mean Score	Accuracy	Precision	Recall	F1-Score
CIC TON IOT	Naive Bayes	0.6085	0.6126	0.82	0.61	0.65
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9742	0.9768	0.93	0.83	0.86
	LogisticRegression	0.9909	0.9937	0.99	0.99	0.99
	Ensemble	0.9926	0.9931	0.99	0.99	0.99
CIC IDS 2017	Naive Bayes	0.8850	0.9039	0.88	0.70	0.75
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9981	0.9986	0.99	0.99	0.99
	LogisticRegression	0.9969	0.9968	0.99	0.99	0.99
	Ensemble	0.9988	0.9991	1.0	1.0	1.0
NSL KDD	Naive Bayes	0.9395	0.9428	0.94	0.90	0.91
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9979	0.9977	1.0	1.0	1.0
	LogisticRegression	0.9977	0.9980	1.0	1.0	1.0
	Ensemble	0.9989	0.9992	1.0	1.0	1.0
KDD CUP 99	Naive Bayes	0.9395	0.9428	0.82	0.61	0.65
	Decision Tree	0.9999	0.9999	1.0	1.0	1.0
	KNeighbors	0.9979	0.9977	0.98	0.98	0.98
	LogisticRegression	0.9977	0.9980	0.99	0.99	0.99
	Ensemble	0.9989	0.9992	0.99	0.99	0.99

Table 11 Anomaly-Based IDS Metrics for Different Datasets

Dataset	Accuracy	Precision	Recall	F1-Score	TP	FP	TN	FN
Anomaly-Based IDS Metrics								
CIC TON IOT	0.9722	0.9722	0.9722	0.9722	13, 398	20	15, 082	2
CIC IDS 2017	0.9722	0.9722	0.9722	0.9722	17, 215	557	19, 525	495
NSL KDD	0.9992	0.9992	0.9992	0.9992	13, 398	20	15, 082	2
KDD Cup 99	0.9722	0.9772	0.9792	0.9992	34	2	36	0

Table 12 Time Comparison for Different Datasets in Seconds

Dataset	Signature-Based IDS	Anomaly Based IDS		Support
		Cluster Creation Time	Classification Time	
CIC-ToN-IoT	14.42	0.16	0.34	28,502
CIC IDS 2017	25.16	75.58	427.07	37,792
NSL KDD	59.31	3.76	193.22	28,502
KDD CUP 99	31.45	0.98	6.41	72

which shows that it can find both known and unknown attacks in IIoT networks.The system's performance evaluations on real-world datasets validate its pragmatic utility and potential for enhancing the security of IIoT environments. Future research focus could be on further refining the anomaly-based IDS framework by exploring a broader spectrum of unsupervised learning techniques and implementing online learning methodologies to adapt to the dynamic IIoT landscape.

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Data Availability No dataset was generated during the research work. The dataset analyzed for this study is generated by the University of Queensland and available on it's website [16].

Declarations

Conflict of interest The authors affirm that they have no known financial or interpersonal conflicts that might have appeared to have influenced the research presented in this paper.

References

1. Tabaa, M.; et al.: Green industrial Internet of Things from a smart industry perspectives. Energy Rep. **6**, 430–446 (2020). https://doi.org/10.1016/j.egyr.2020.09.022
2. Yadav, G.; Paul, K.: Architecture and security of SCADA systems: a review. Int. J. Crit. Infrastruct. Prot. **34**, 100433 (2021). https://doi.org/10.1016/j.ijcip.2021.100433
3. Nuaimi, M.; Fourati, L.C.; Hamed, B.B.: Intelligent approaches toward intrusion detection systems for industrial Internet of Things: a systematic comprehensive review. J. Netw. Comput. Appl. **215**, 103637 (2023)
4. Gupta, P.; et al.: Industrial internet of things in intelligent manufacturing: a review, approaches, opportunities, open challenges, and future directions. Int. J. Interact. Des. Manuf. (IJIDeM) (2022). https://doi.org/10.1007/s12008-022-01075-w
5. Saurabh, K.; et al.: TMAP: A Threat Modeling and Attack Path Analysis Framework for Industrial IoT Systems (A Case Study of IoM and IoP). (2023). arXiv:2312.15319 [cs.CR].
6. Pourrahmani, H.; et al.: The applications of Internet of Things in the automotive industry: a review of the batteries, fuel cells, and engines. Internet Things **19**, 100579 (2022)
7. Rejeb, A.; et al.: Unleashing the power of internet of things and blockchain: a comprehensive analysis and future directions. Internet Things Cyber Phys. Syst. **4**, 1–18 (2023)
8. Shamshirband, S.; et al.: Computational intelligence intrusion detection techniques in mobile cloud computing environments: review, taxonomy, and open research issues. J. Inf. Secur. Appl. **55**, 102582 (2020)
9. Sethi, K.; et al.: Attention based multi-agent intrusion detection systems using reinforcement learning. J. Inf. Secur. Appl. **61**, 102923 (2021)
10. Jin, M.; et al.: A Survey on Graph Neural Networks for Time Series: Forecasting, Classification, Imputation, and Anomaly Detection (2023). arXiv preprint arXiv:2307.03759
11. Singh, U.; et al.: Anomaly classification to enable self-healing in cyber physical systems using process mining. In: Sellmann, M., Tierney, K. (eds.), et al., Learning and Intelligent Optimization, pp. 1–15. Springer, Cham (2023)
12. Zahoora, U.; et al.: Zero-day ransomware attack detection using deep contractive autoencoder and voting based ensemble classifier. Appl. Intell. **52**(12), 13941–13960 (2022)
13. Saurabh, K.; et al.: Lbdmids: LSTM based deep learning model for intrusion detection systems for IOT networks. In: IEEE World AI IoT Congress (AIIoT), pp. 753–759. IEEE (2022)
14. Saurabh, K.; et al.: NFDLM: a lightweight network flow based deep learning model for DDoS attack detection in IoT domains. In: IEEE World AI IoT Congress (AIIoT), vol. 2022, pp. 736–742. IEEE (2022)
15. Saurabh, K.; et al.: GANIBOT: a network flow based semi supervised generative adversarial networks model for IoT botnets detection. In: 2022 IEEE International Conference on Omni-layer Intelligent Systems (COINS), pp. 1–5. IEEE (2022)
16. Saurabh, K.; et al.: MLAPS: a machine learning based second line of defense for attack prevention in IoT network. In: IEEE 19th India Council International Conference (INDICON), vol. 2022, pp. 1–6. IEEE (2022)
17. Halabi, T.; et al.: Protecting the internet of vehicles against advanced persistent threats: a Bayesian Stackelberg game. IEEE Trans. Reliab. **70**(3), 970–985 (2021)
18. Khraisat, A.; et al.: Survey of intrusion detection systems: techniques, datasets and challenges. Cybersecurity **2**(1), 1–22 (2019)
19. Markevych, M.; Dawson, M.: A review of enhancing intrusion detection systems for cybersecurity using artificial intelligence (AI). In: International conference Knowledge-Based Organization, vol. 29, no. 3, pp. 30–37 (2023)
20. Soltani, M.; et al.: An adaptable deep learning-based Intrusion Detection System to zero-day attacks. J. Inf. Secur. Appl. **76**, 103516 (2023)
21. Lin, H.; Wu, C.; Masdari, M.: A comprehensive survey of network traffic anomalies and DDoS attacks detection schemes using fuzzy techniques. Comput. Electr. Eng. **104**, 108466 (2022)
22. Zhang, Z.; et al.: Unknown attack detection based on zero-shot learning. IEEE Access **8**, 193981–193991 (2020). https://doi.org/10.1109/ACCESS.2020.3033494
23. Hindy, H.; et al.: Utilising deep learning techniques for effective zero-day attack detection. Electronics **9**(10), 1684 (2020)
24. Li, Z.; et al. Zero-shot learning for intrusion detection via attribute representation. In: Neural Information Processing: 26th International Conference, ICONIP 2019, Sydney, NSW, Australia, December 12–15, 2019, Proceedings, Part I 26, pp. 352–364. Springer (2019)
25. Kumar, V.; Sinha, D.: A robust intelligent zero-day cyber-attack detection technique. Complex Intell. Syst. **7**(5), 2211–2234 (2021)
26. Sabeel, U.; et al.: Building an intrusion detection system to detect atypical cyberattack flows. IEEE Access **9**, 94352–94370 (2021)
27. Mokhtari, S.; et al.: A machine learning approach for anomaly detection in industrial control systems based on measurement data. Electronics **10**(4), 407 (2021)
28. Khan, A.S.; et al.: A spectrogram image-based network anomaly detection system using deep convolutional neural network. IEEE Access **9**, 87079–87093 (2021)
29. Hwang, R.-H.; et al.: An unsupervised deep learning model for early network traffic anomaly detection. IEEE Access **8**, 30387–30399 (2020)
30. Feng, F.; et al.: Anomaly detection in ad-hoc networks based on deep learning model: a plug and play device. Ad Hoc Netw. **84**, 82–89 (2019). https://doi.org/10.1016/j.adhoc.2018.09.014
31. Abdalgawad, N.; et al.: Generative deep learning to detect cyberattacks for the IoT-23 dataset. IEEE Access **10**, 6430–6441 (2022). https://doi.org/10.1109/ACCESS.2021.3140015
32. Sarhan, M.; Layeghy, S.; Portmann, M.: Evaluating standard feature sets towards increased generalisability and explainability of ML-based network intrusion detection. Big Data Res. **30**, 100359 (2022). https://doi.org/10.1016/j.bdr.2022.100359
33. Maini, S.; Aggarwal, A.K.: Camera Position Estimation using 2D Image Dataset (2018). https://api.semanticscholar.org/CorpusID:225091809
34. Ravipati, R.D.; Abualkibash, M.: Intrusion detection system classification using different machine learning algorithms on KDD-99 and NSL-KDD datasets—a review paper. SSRN Electron. J. **11**, 16 (2019)

35. Tavallaee, M.; et al.: A detailed analysis of the KDD CUP 99 data set. In: 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications, pp. 1–6 (2009). https://doi.org/10.1109/CISDA.2009.5356528.
36. Sharafaldin, I.; Lashkari, A.H.; Ghorbani, A.A.: Toward generating a new intrusion detection dataset and intrusion traffic characterization. In: International Conference on Information Systems Security and Privacy (2018). https://api.semanticscholar.org/CorpusID:4707749
37. Singh, U.; et al.: Hyperparameter tuning for LSTM and ARIMA time series model: a comparative study. In: 2023 IEEE 4th Annual Flagship India Council International Subsections Conference (INDISCON), pp. 1–6 (2023). https://doi.org/10.1109/INDISCON58499.2023.10270325
38. Sun, L.; et al.: Feature selection using symmetric uncertainty and hybrid optimization for high-dimensional data. Int. J. Mach. Learn. Cybern. **14**, 1–22 (2023)
39. Yu, L.; Liu, H.: Efficiently handling feature redundancy in high-dimensional data. In: Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. KDD '03, pp. 685–690. Association for Computing Machinery, Washington (2003). ISBN: 1581137370. https://doi.org/10.1145/956750.956840.
40. Lashkari, A.H.; et al.: Characterization of tor traffic using time based features. In: Proceedings of the 3rd International Conference on Information Systems Security and Privacy, ICISSP, INSTICC, vol. 1, pp. 253–262. SciTePress (2017). ISBN: 978-989-758-209-7. https://doi.org/10.5220/0006105602530262
41. Shetty, G.; Raman, K.K.: Performance analysis of a snort-based intrusion detection system for wireless sensor networks. Journal homepage: www.ijrpr.com ISSN 2582, p. 7421.
42. Zhang, C.; et al.: Unsupervised anomaly detection based on deep autoencoding and clustering. Secur. Commun. Netw. **2021**, 1–8 (2021)
43. Yang, L.; Shami, A.: On hyperparameter optimization of machine learning algorithms: theory and practice. Neurocomputing **415**, 295–316 (2020). https://doi.org/10.1016/j.neucom.2020.07.061
44. Ghahramani, Z.: Probabilistic machine learning and artificial intelligence. Nature **521**(7553), 452–459 (2015)
45. Mohammed, M.; et al.: Using stacking ensemble for microarray-based cancer classification. In: 2018 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE), pp. 1–8. IEEE (2018)
46. Na, S.; Xumin, L.; Yong, G.: Research on k-means clustering algorithm: an improved k-means clustering algorithm. In: Third International Symposium on Intelligent Information Technology and Security Informatics, pp. 63–67. IEEE (2010)
47. Feizollah, A.; et al.: Comparative study of k-means and mini batch k-means clustering algorithms in android malware detection using network traffic analysis. In: 2014 International Symposium on Biometrics and Security Technologies (ISBAST), pp. 193–197 (2014). https://doi.org/10.1109/ISBAST.2014.7013120
48. Vikram, A.; Mohana: anomaly detection in network traffic using unsupervised machine learning approach. In: 2020 5th International Conference on Communication and Electronics Systems (ICCES), pp. 476–479 (2020). https://doi.org/10.1109/ICCES48766.2020.9137987.

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